



Computational Intelligence Research Foundation

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Analyzing ARP Request

Feb 20, 2025

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Department: BCA

Year of Study: 2024 - 2027





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Dr Doreen Hephzibah Miriam M.E., Ph.D., is the founder and Director of CIRF.

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- Support higher education institutions to establish Innovation Ecosystem

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- Collaborate with Institutions, research and development organizations and Industries.
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CIRF collaborates with various institutions, research and development organizations, and industries to foster innovation and development. The organization also aims to serve the community at large through its initiatives and projects.

Contact Information

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CIRF is committed to driving innovation and research to support national development and address complex problems through advanced computational intelligence techniques.



TO WHOMSOEVER IT MAY CONCERN

This is to certify that Shreya V of batch : 2024 - 2027, Department of BCA, Shrimathi Devkunvar Nanalal Bhatt Vaishnav College for Women has done their Value-added course on Cybersecurity at Computational Intelligence Research Foundation (CIRF), Chennai from February 5,2025 to February 20,2025.

She had worked on a project titled, **Analyzing ARP Request** . This Project was aimed to analyse the Address Resolution Protocol request using wireshark.

During their **Value-added course on Cybersecurity**, she demonstrated a strong enthusiasm for learning new technologies and tools and exhibited exceptional creativity in exploring and applying cybersecurity concepts. The above project has been developed under my guidance.

We wish all the best for his future endeavors

Dr D Doreen Hephzibah Miriam M E., Ph D.,
Director & Founder,
Computational Intelligence Research Foundation (CIRF),
Chennai 600 023.

Internship attendance position of Shreya V- BCA 1st Year

We hereby certify that the attendance position of Shreya V- BCA 1st Year for the period of February 5,2025 to February 20,2025 as follows;

ATTENDANCE SHEET

ATTENDANCE SHEET

S.No	Date	Hours	Topics	Time In	Time Out
1	05.02.2025	3	• Introduction to Cyber Security	9:30 am	12:30 pm
2	06.02.2025	3	• Cyber Security Tools and Introduction to Wireshark	9:30 am	12:30 pm
3	07.02.2025	3	• Wireshark Exercises • Phishing and it's types	9:30 am	12:30 pm
4	10.02.2025	3	• Introduction to BurpSuite	9:30 am	12:30 pm
5	12.02.2025	3	• Working with BurpSuite • Types of Cyber attacks	9:30 am	12:30 pm
6	13.02.2025	3	• Working with BurpSuite • Introduction to Types of Cyber Crimes	9:30 am	12:30 pm
7	14.02.2025	3	• Types of Cyber Crimes	9:30 am	12:30 pm
8	17.02.2025	3	• Introduction to Cyber Laws • Cyber Laws and it's Punishments	9:30 am	12:30 pm
9	18.02.2025	3	Working with the Final Project Presentation	9:30 am	12:30 pm
10	19.02.2025	3	Final Project Presentation	9:30 am	12:30 pm
11	20.02.2025	3	Final Project Presentation	9:30 am	12:30 pm

WORK LOG

S.NO.	DATE	PARTICULARS	TIME
1.	05.02.2025 - 07.02.2025	Introduction to Cyber Security, Cyber Security Tools, Wireshark and Exercises	9:30 - 12:30
2.	10.02.2025 - 14.02.2025	Introduction to BurpSuite, Working with Exercises - BurpSuite, Cyber Crimes and it's types	9:30 - 12:30
3.	17.02.2025 - 18.02.2025	Cyber law and it's Punishments	9:30 - 12:30
4.	19.02.2025 - 20.02.2025	Final Project Presentation	9:30 - 12:30

ABSTRACT

This document provides an in-depth analysis of ARP (Address Resolution Protocol) and its role in network security, focusing on detecting ARP spoofing using Wireshark. It covers the installation, configuration, and application of Wireshark for capturing and analyzing ARP traffic. The paper explores ARP vulnerabilities, prevention techniques, and real-world case studies, including mitigating ARP spoofing in corporate networks. It also discusses cyber laws related to ARP attacks in India and highlights key findings from ARP traffic analysis, such as network inefficiencies and security threats. The research emphasizes Wireshark's effectiveness in identifying and managing ARP-related issues.

1 INTRODUCTION

Wireshark is a free and open-source packet analyzer. It is used for network troubleshooting, analysis, software and communications protocol development, and education. Packet is the name given to a discrete unit of data in a typical Ethernet network.

Wireshark is the most often-used packet sniffer in the world. Like any other packet sniffer, Wireshark does three things:

- Packet Capture
- Filtering
- Visualization

Address Resolution Protocol (ARP) The Address Resolution Protocol is a layer 2 protocol used to map MAC addresses to IP addresses.

1.1 Goals

- **Goal1:**Capture the packets and filter the arp packets
- **Goal 2:** Analyse the arp request packets

1.2 Objectives

1. The primary objective of analyzing ARP requests using Wireshark is to understand the mapping between IP addresses (Layer 3) and MAC addresses (Layer 2) on a

network, allowing you to identify active devices, troubleshoot connectivity issues, and detect potential security threats like ARP spoofing by examining the source and destination IP and MAC addresses within captured ARP packets.

- Key objectives include:

Identifying network devices

Troubleshooting connectivity problems

Detecting ARP spoofing attacks

2 LITERATURE REVIEW - 5 PAPERS

S. No	Title of the Paper	Author Name	Proposed Solution	Journal Conference Name	Year of Publication
1.	Security Analysis and Simulation Experiment of ARP Protocol Vulnerability	Tong, et.al	This paper examines ARP protocol vulnerabilities through simulation of ARP spoofing and flooding attacks, using Wireshark for analysis, and proposes preventive measures.	International Core Journal of Engineering	2024
2.	A Strategy to Mitigate ARP Spoofing Attacks on Hypervisors	Mohamed, et.al	A study found virtual machines in a multi-tenant cloud vulnerable to ARP spoofing, with those running multiple OSes being more susceptible. A Kali Linux Python script mitigated the attacks, but no hypervisor was fully immune.	2024 Fourth ICAECT	2024
3.	Forensic Analysis of an IoT ARP Spoofing Attack	Friedl, et.al	ARP spoofing in IoT enables hackers to steal traffic, as shown in a case of industrial espionage, emphasizing the need for forensic analysis to detect such advanced attacks.	2024 12th ISDFS	2024
4.	An Effective Prevention	Suethanuw	This paper proposes	ECTI-CIT	2024

	Approach against ARP Cache Poisoning Attacks in MikroTik-based Networks	ong, et.al	Gateway-controlled ARP (GCA) to prevent ARP cache poisoning in MikroTik-based RoaS networks by forwarding ARP messages to the router for inspection, effectively blocking spoofed packets.		
5.	Arp spoofing detection using machine learning classifiers: an experimental study	Majumder, et.al	This study develops a real-time ARP spoofing detection approach, optimizing ML classifiers for continuous IP/MAC verification, achieving 99.26% F1-Score with CNN for real-time detection.	Knowledge and Information Systems	2025

2.1 INSTALLATION AND SETUP

Install Wireshark:

- Download and install Wireshark from Wireshark's official website.
- Follow the installation process for your operating system (Windows, Linux, macOS).

Set Up Network Interfaces:

- Ensure Wireshark is running with the necessary privileges to capture network traffic.
- Select the correct network interface (Ethernet, Wi-Fi, etc.) that you want to monitor.

3 ANALYSING CAPTURED DATA

- Once capturing is complete
- You will find many packets captured and in that

1. Light blue:
❖ UDP Traffic
2. Light purple:
❖ TCP traffic

3. Black:
 - ❖ Packets with error
4. Analyze ARP Requests
 - ❖ Look for normal ARP requests, which ask for MAC addresses (e.g., "Who has 192.168.1.1? Tell 192.168.1.100").

3.1 Applying Filters / Intercepting & Modifying Requests

- 1. Applying Filters for ARP Traffic
 Filtering ARP packets helps focus on specific network interactions.
 Wireshark Filters
 Capture Filter: To capture only ARP packets
 Display Filter: To analyze ARP traffic after capture

4. Cyber Laws and Punishments

- ARP spoofing or attacks that modify network traffic are illegal in India under cybercrime laws. The IT Act, 2000, and Indian Penal Code (IPC) cover such offenses.
- Key Laws & Punishments:
 1. Unauthorized Access & Hacking (IT Act, Sec 43, 66)

Modifying network traffic without permission is illegal.

Punishment: Up to 3 years jail or ₹5 lakh fine.

2. Identity Theft & Fraud (IT Act, Sec 66B, 66C, 66D)

Using ARP spoofing to steal data or impersonate someone is punishable.

Punishment: Up to 3 years jail and fines.

3. Privacy Violation (IT Act, Sec 72)

Stealing private data via ARP attacks is a crime.

Punishment: Up to 2 years jail or fine.

4. Financial Fraud (IPC Sec 420)

If ARP spoofing is used for scams, fraud laws apply.

Punishment: Up to 7 years jail.

5 Case Study of Wireshark

Case Title: Detecting and Mitigating ARP Spoofing in a Corporate Network

- Background

A mid-sized company with 200+ employees started experiencing intermittent network slowdowns and unauthorized access attempts. The IT team suspected a potential security threat but lacked concrete evidence. They decided to use Wireshark to analyze ARP traffic and identify anomalies.

- Objectives

Capture and analyze ARP traffic to detect unusual patterns.

Identify signs of ARP spoofing or unauthorized devices.

Implement security measures based on findings.

- Methodology

1. Capturing ARP Traffic Using Wireshark

The IT team ran Wireshark on a mirrored switch port to capture all ARP requests and replies.

Filter applied:

The capture was run for 24 hours during business operations.

2. Key Observations from Wireshark Analysis

(a) High Frequency of ARP Requests

The network exhibited an unusually high volume of ARP requests.

- Certain IP addresses were repeatedly requesting MAC addresses, which could indicate an attack or misconfiguration.

b) Duplicate ARP Replies from Different MAC Addresses

Wireshark revealed multiple ARP replies for the same IP address but from different MAC addresses, indicating possible ARP spoofing.

(c) Presence of Gratuitous ARP (GARP) Packets

A rogue device was sending unsolicited ARP replies, attempting to redirect traffic.

(d) Network Slowdown Correlation

The periods of high ARP traffic aligned with users reporting slow network speeds and dropped connections.

- Findings and Analysis

ARP Spoofing Detected: A rogue device was impersonating the default gateway (192.168.1.1), attempting to intercept traffic (Man-in-the-Middle attack).

Potential Data Interception: The attacker could have been sniffing sensitive

employee credentials.

Network Performance Issues: The excess ARP traffic led to network congestion.

- Mitigation Measures Implemented

Implemented Dynamic ARP Inspection (DAI)

The IT team enabled DAI on the core switch, preventing unauthorized ARP replies.

Configured Static ARP Entries for Critical Devices

Critical network devices (gateway, DNS server) were assigned static ARP entries to prevent spoofing.

Used ARPWatch for Continuous Monitoring

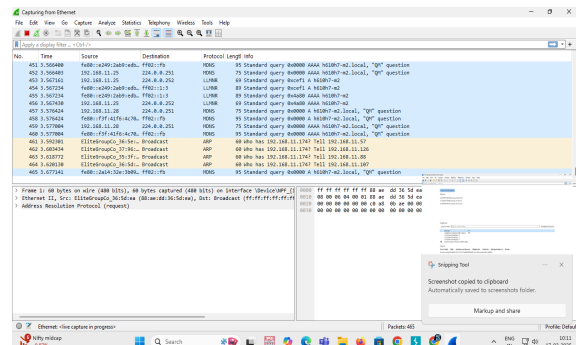
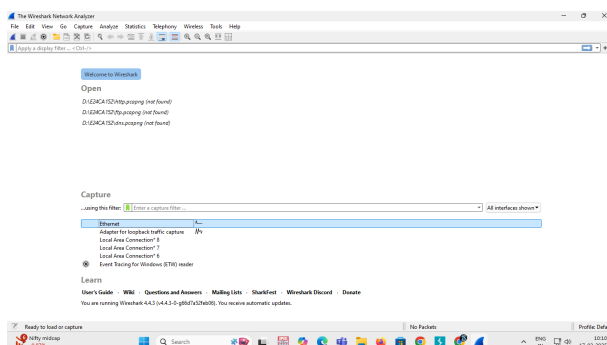
- Results & Impact

Reduced ARP Traffic: After implementing security measures, ARP request volume normalized.

Prevented Further Attacks: No more unauthorized ARP replies were observed.

Enhanced Security Monitoring: Real-time alerts were set up to detect future ARP anomalies.

6. Screenshots



The top two screenshots show Wireshark packet captures of ARP requests. The first screenshot shows a list of ARP requests from various hosts to the broadcast address 192.168.1.255. The second screenshot shows a detailed view of an ARP request packet, including the Ethernet II header, Internet Protocol header, and ARP request payload.

The bottom screenshot shows a Windows Command Prompt window with the following output:

```

C:\Users\acer>arp -a
arp -a is not recognized as an internal or external command,
operable program or batch file.

C:\Users\acer>arp -a

Interface: 192.168.11.24 --- 0x4
Internet Address      Physical Address      Type
192.168.11.180         88-2a-3d-4e-18-a0     dynamic
192.168.11.255         ff-ff-ff-ff-ff-ff     static
224.0.0.22            01-00-5e-00-00-16     static
224.0.0.251           01-00-5e-00-00-fb     static
224.0.0.252           01-00-5e-00-00-fc     static
239.255.255.250       01-00-5e-7f-ff-fa     static
  
```

7. Challenges and Limitations

1. High Volume of ARP Traffic

- Large networks generate excessive ARP requests, making it difficult to filter relevant data.
- Broadcast-heavy ARP traffic can clutter packet captures, leading to analysis fatigue.

2. Short ARP Cache Lifespan

- ARP cache timeout varies across devices, leading to frequent re-resolution of addresses.
- Short cache durations increase the volume of ARP requests, making long-term trend analysis challenging.

3. Limited Visibility in Switched Networks

- Wireshark only captures packets visible to the interface it is running on.
- In a switched environment, ARP requests from other VLANs or network segments remain unseen unless configured with port mirroring or network taps.

4. Incomplete ARP Responses

- Some ARP requests go unanswered, making it difficult to determine whether a device is offline, misconfigured, or blocking responses.
- DHCP-based dynamic addressing can cause inconsistencies in request-response tracking.

8. Key Findings and Insights

Frequent ARP Requests:

- A significant number of ARP requests were observed, indicating active device discovery and address resolution on the network.
- High-frequency ARP traffic may suggest network congestion or inefficient ARP handling.

Broadcast Traffic Dominance:

- Most ARP requests are broadcasted (sent to MAC address FF:FF:FF:FF:FF:FF), which can lead to increased network load if unmanaged.

Unresolved ARP Requests:

- Some ARP requests remain unanswered, potentially indicating inactive devices, misconfigured hosts, or security concerns like IP conflicts.

9. Conclusion

Analyzing ARP requests using Wireshark provides valuable insights into network activity, security threats, and performance issues. The findings from the analysis help identify network inefficiencies, unauthorized devices, and potential cyber threats such as ARP spoofing and Man-in-the-Middle attacks. Wireshark proved to be a critical tool in identifying ARP spoofing and network inefficiencies. The insights gained helped the company secure its network and improve performance through better ARP management.

10 References

1. Tong, H., Zhou, G., & Liu, S. (2024). Security Analysis and Simulation Experiment of ARP Protocol Vulnerability. *International Core Journal of Engineering*, 10(1), 288-294.
2. Friedl, S., & Pernul, G. (2024, April). Forensic Analysis of an IoT ARP Spoofing Attack. In *2024 12th International Symposium on Digital Forensics and Security (ISDFS)* (pp. 1-7). IEEE.

3. Mohamed, Y. A., Hashim, M., & Bashir, M. (2024, January). A Strategy to Mitigate ARP Spoofing Attacks on Hypervisors. In 2024 Fourth International Conference on Advances in Electrical, Computing, Communication and Sustainable Technologies (ICAECT) (pp. 1-8). IEEE.
4. Suethanuwong, E. (2025). An Effective Prevention Approach against ARP Cache Poisoning Attacks in MikroTik-based Networks. ECTI Transactions on Computer and Information Technology (ECTI-CIT), 19(1), 1-12.
5. Majumder, S., Deb Barma, M. K., & Saha, A. (2025). Arp spoofing detection using machine learning classifiers: an experimental study. Knowledge and Information Systems, 67(1), 727-766.